

VPN-LAN Service Access — Plan Summary

Revision: 2 **Last modified:** 2026-07-01T17:05:00Z **Status:** Companion summary of [PLAN.md](#) (§11.4.56 two-audience). Rev 2 adds the operator's **bidirectional** mandate (2026-07-01): exposure must work fully **both ways** (VPN hosts can also reach exposed proxy-side services — needed by FTP active, NFS lock callbacks, Cast callbacks, adb reverse, etc.), governed by a **default-deny + allowlist** ingress surface (the mirror of the egress SSRF floor) and covered by all supported test types — see Phase 12 + `bidirectional_exposure.md`.

Page 1 — For the operator / stakeholders (plain language)

We are building the ability for `helix_proxy` to reach and use **all the mainstream services running on the far side of the VPN** — file shares (Windows/SMB and Unix/NFS), file transfer (FTP, SFTP, WebDAV), email (IMAP/SMTP/POP3), device discovery (Bonjour, UPnP), casting (Chromecast), and Android devices over ADB (including flashing). The VPN connection itself is driven by your existing `sword_toolkit` scripts — but `helix_proxy` never hardcodes them: it reads the script location from a setting you provide in a private `.env` file (a tracked `.env.example` shows exactly how to point it at `sword_toolkit`).

How it works, in one sentence: normal one-to-one services (files, email, ADB) are simply *routed* over the VPN; web-style services (WebDAV) already work through the existing proxy; device *discovery* (which uses network-wide "shout" messages that don't cross the VPN) needs a small helper deployed on the far side; and one thing — **Miracast** — is **physically impossible** over this kind of VPN (it's a direct radio link, not an internet protocol), so we honestly say so and offer **Chromecast** as the working alternative.

What we will prove, with saved evidence, no faking: every protocol gets a real round-trip test (write a file and read it back byte-for-byte, list a real mailbox, connect to a real Android

device). When the live VPN isn't connected (which needs your credentials), the tests **honestly skip** instead of pretending to pass.

What needs you (parked until you decide): the live VPN connection (secrets + Mullvad + admin rights), deploying the discovery helper on a remote host, and flashing any real device — each of these we will **ask you about with clear options** before doing anything, and we will **never** change sword_toolkit or any remote host without asking first.

Bottom line: the full architecture is designed and security-checked; we can build and prove all the "logic" parts autonomously right now, and the "live" parts switch on the moment you provide the VPN connection. Nothing is overstated, nothing is faked.

Page 2 — For software engineers

L3-routed VPN (WireGuard+L2TP/PPP, 10.0.0.0/8, sword host 10.6.100.221) ⇒ **route unicast, proxy HTTP-shaped, reflect multicast, structurally-reject L2**. Bridge is env-var-decoupled (§11.4.28): HELIX_SWORD_DIR / HELIX_BRIDGE_{CONNECT,DISCONNECT,HEALTH} / HELIX_BRIDGE_{SUBNET,HOST}, real values in gitignored .env, shape in tracked .env.example (§11.4.30/§11.4.77). SSRF reconciliation (§11.4.120): keep RFC1918/metadata floor, add narrow HELIX_BRIDGE_SUBNET carve-out (Dante first-match above internal-deny), S1/S3/S4 guards stay GREEN + paired §1.1 teeth; open-relay guard on email (§4.3).

Phase	Scope	Primitive	Autonomous now?
0	Bridge scaffold + sword-doctor preflight	env-var contract + honest SKIP	YES
1	L3 routed gateway + SSRF allowlist reconciliation	route + Dante/Squid carve-out	logic YES (local stub)
2	SMB/CIFS/NMB + NFS	route (mount round-trip + sha256)	live-gated
3	FTP/FTPS/SFTP + WebDAV	route + passive range; WebDAV via Squid	live-gated (WebDAV logic YES)
4	IMAP/IMAPS/SMTP-submission/POP3S	route/CONNECT +	live-gated

Phase	Scope	Primitive	Autonomous now?
		open-relay guard	
5	mDNS/SSDP/WS-Discovery/DNS-SD reflector	remote-side reflector (containers §11.4.76)	operator-gated deploy
6	Chromecast/DIAL	reflect discovery + route control (8008/8009)	live-gated
7	ADB (access/debug/connect/flash)	route 5555 + adb server + usbip fastboot	live+device-gated
8	Miracast	structurally-impossible §11.4.112 + Cast alt	YES (verdict)
9/10/11	docs / containerize / §11.4.169 full test-type coverage + Challenges + HelixQA	—	continuous

Critical path: Phase 0 → Phase 1 (routed gateway + reconciled SSRF) gates all unicast; Phase 5 → Phase 6 for Cast; Phase 8 independent. Evidence under qa-results/vpn_lan/<phase>/<ts>; bridge-down ⇒ §11.4.3 honest SKIP (never fail-open, §11.4.69). Deep-research citations §11 (§11.4.150). Composes §11.4.28/ §11.4.58/§11.4.66/ §11.4.76/§11.4.101/§11.4.112/§11.4.115/ §11.4.120/§11.4.122/§11.4.133/§11.4.167/§11.4.169.